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A place to talk about Modular Synthesizers. And Cats.

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303 circuit modelling

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Re: 303 circuit modelling

by [toneburst](#) » Wed Jun 11, 2025 6:13 am

Interesting... I wonder if Open303 emulates that. I haven't spent much time looking at the osc-related classes.

Looking again, there does seem to be some attempt to make the square waveform not-quite-square, but I can't tell on my cursory look if it emulates the frequency-related shape behaviour above.

Re: 303 circuit modelling

by [guest](#) » Wed Jun 11, 2025 11:09 am

theres a lot of code in there, and im having trouble locating where the square is generated. can you point me to it?

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Re: 303 circuit modelling

by [sphet](#) » Wed Jun 11, 2025 11:52 pm

guest,

It looks like maybe it is here:

<https://github.com/RobinSchmidt/Open303...eTable.cpp>

CODE: SELECT ALL

```
void MipMappedWaveTable::fillWithSquare303()
{
    // generate the saw-wave:
    int N = tableLength;
    double k = 0.5;
    int N1 = clip(roundToInt(k*(N-1)), 1, N-1);
    int N2 = N-N1;
    double s1 = 1.0 / (N1-1);
    double s2 = 1.0 / N2;
    for(int n=0; n<N1; n++)
        prototypeTable[n] = s1*n;
    for(int n=N1; n<N; n++)
        prototypeTable[n] = -1.0 + s2*(n-N1);
}
```

Re: 303 circuit modelling

by [guest](#) » Thu Jun 12, 2025 12:52 am

yeah, i dont think that would make a very accurate square wave. its a tanh with offset. assuming those parameters are varied and square waves are generated per note, then the pwm shift could be modeled, and the 'squareness' could be modified, on a note by note basis, but the shape wouldnt be very accurate except at the higher frequencies.

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Re: 303 circuit modelling

by [toneburst](#) » Thu Jun 12, 2025 9:21 am

Interesting... makes me wonder how accurately the other elements are modelled.

Re: 303 circuit modelling

by [Houly](#) » Sun Jun 15, 2025 6:56 am

Hello,
Congrat Guest for your impressive work ! I'm very interested in your project especially the way for modeling the loop. Have you planned any documentation on this subject ?

Re: 303 circuit modelling

by [guest](#) » Sun Jun 15, 2025 12:13 pm

its all up on the wiki right now, but not well documented:

<http://wiki.openmusiclabs.com/wiki/Digix0x>openmusiclabs.com[toneburst](#)
Super Deluxe Wiggler

Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London

[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

[sphet](#)
Common Wiggler

Posts: 91
Joined: Tue Aug 21, 2018 10:24 pm

[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

[toneburst](#)
Super Deluxe Wiggler

Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London

[Houly](#)
Learning to Wiggle

Posts: 4
Joined: Thu Jun 12, 2025 11:57 am



Re: 303 circuit modelling

by Houly » Tue Jun 17, 2025 8:42 am

Thank you Guest for the spreadsheet

Re: 303 circuit modelling

by connortreacy » Wed Jul 09, 2025 9:46 am

This is such an exciting project! I built a proto board version last night and it sounds great. I'd love to help with this project any way I can. Did you have any plans for what's next?

<https://github.com/connortreacy/digix0x>

Re: 303 circuit modelling

by guest » Fri Jul 11, 2025 9:01 pm

connortreacy wrote:

Wed Jul 09, 2025 9:46 am

This is such an exciting project! I built a proto board version last night and it sounds great. I'd love to help with this project any way I can. Did you have any plans for what's next?

this is great news, and exactly what i was hoping for. ive been travelling for a while, but will be getting back to this project shortly. i need to make the bandlimited wavetables, and then the audio section will 99% done. then i was thinking of making a small sequencer, or maybe a midi implementation for it. any help would be appreciated.

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Re: 303 circuit modelling

by connortreacy » Tue Jul 15, 2025 6:22 am

“

i was thinking of making a small sequencer, or maybe a midi implementation for it

I'm on it! I started that at the weekend actually. I've got it syncing to the midi clock over USB MIDI and was gonna start implementing midi note support this week.

Hope you don't mind, but I've put up your code on a private github repo. Message me your github handle and I'll add you as a collaborator, or if you'd prefer I can transfer the repo to you and fork it myself.

<https://github.com/connortreacy/digix0x>

Re: 303 circuit modelling

by guest » Tue Jul 15, 2025 10:43 am

awesome work. I dont have a github, so using your repo is fine. also, feel free to make it public, the more the merrier! hopefully i can get back to the project in a few days. the other thing id like to do is rewrite the code for 48kps, so that it can be used with an audio DAC if desired.

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Re: 303 circuit modelling

by connortreacy » Tue Jul 15, 2025 6:41 pm

Love it! Here's the repo: <https://github.com/connortreacy/digix0x>. You should set up a github account and become a collaborator. And let me know if you want any clearer attribution, or a licence file or anything.

I've added all the midi stuff (sync only for now) in a branch here <https://github.com/connortreacy/digix0x/tree/midi>, and I'll try adding a few more things on other branches as I go, but any major changes to the audio core should prob happen on main.

<https://github.com/connortreacy/digix0x>

Re: 303 circuit modelling

by connortreacy » Thu Jul 17, 2025 4:03 am

Went on a bit of a fool's errand last night soldering in a PCM5102 DAC (I had one spare), only to find the M4 Express doesn't support i2s! 🤦

Any specific reason for choosing ItsyBitsy M4 as the platform for this one? I think it should be possible to get it running on a bunch of different platforms easily enough, or a different DAC, but this isn't something I've got much experience in.

<https://github.com/connortreacy/digix0x>

Re: 303 circuit modelling

by guest » Thu Jul 17, 2025 10:03 am

im using the feather m4 for my prototyping, which has i2s, but i havent implemented it yet. sometimes spi can be hacked to replicated i2s.

i just noticed i have no idea where my hardware is. i had to put the project on hold for a few months due to other things needing priority, and now i cant find it. im sure its around here someplace.

guest
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

Houly
Learning to Wiggle

Posts: 4
Joined: Thu Jun 12, 2025 11:57 am

connortreacy
Learning to Wiggle

Posts: 7
Joined: Wed Jul 09, 2025 8:23 am
Location: London
Contact: [📧](#)



guest
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

connortreacy
Learning to Wiggle

Posts: 7
Joined: Wed Jul 09, 2025 8:23 am
Location: London
Contact: [📧](#)



guest
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

connortreacy
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Location: London
Contact: [📧](#)

connortreacy
Learning to Wiggle

Posts: 7
Joined: Wed Jul 09, 2025 8:23 am
Location: London
Contact: [📧](#)



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Re: 303 circuit modelling

by **connortreacy** » Thu Jul 17, 2025 5:09 pm

Hope you find it soon! I got midi note playback working, albeit it's pretty rough! <https://github.com/connortreacy/digix0x...igix0x.ino>

<https://github.com/connortreacy/digix0x>

Re: 303 circuit modelling

by **guest** » Mon Jul 21, 2025 9:35 pm

sounding good. are the notes shutting off in your demo, or are they constantly playing (sliding).

also, found my board!

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Re: 303 circuit modelling

by **connortreacy** » Tue Jul 22, 2025 5:20 am

They're shutting off, but the sliding isn't working correctly and I ran out of time to investigate. I'll take another look this week, but do take a look at the midi fork on my github and see if you have any ideas!

<https://github.com/connortreacy/digix0x>

Re: 303 circuit modelling

by **guest** » Sat Aug 02, 2025 10:43 pm

well, im finally making some progress here. it took me a long time to figure out how to sample the saw and square waves, or whether i should sample them at all. i could algorithmically generate them, digitally filter and create samples. this would have the advantage of being noise free, frequency accurate, and easy to make at any sample rate i want. the downside is that i would either have to create them at some ridiculous sample rate before filtering (as a mathematically accurate sawtooth has infinite harmonics, which would alias before the filter even hit it) or apply something like polyblep after waveform generation. the other downside is that the square would never be perfectly true to the original. it would probably be close enough, though. the advantage of sampling the synth is that the waveforms are accurate, right on down to the noise and 60Hz modulation (unfortunately), and they can be pre-filtered in analog to deal with aliasing issues. the downside is that the number of sample points per waveform drops off rather quickly, so i would need to come up with an interpolation algorithm that made some sense.

i decided to sample the synth, and spent all day today collecting samples. i put them up on the wiki if anyone wants to use them:

<http://wiki.openmusiclabs.com/wiki/Digix0x>

these are not the final lookup tables, but rather just raw audio capture of the synth for both square and saw across the full frequency range. i collected midi notes 12-76, which is almost the full range of the synth. it covers playing the lowest note and then turning the tune knob all the way down (basically another octave down), and playing the highest note and adding octave shift to a track. i have 4 semitones of tune shift past that top note, but not the full octave youd usually get. there were some limitations in how easy/hard it is to get the synth to make those notes.

the output was taken right from the waveform switch, and passed through a 4 pole bessell filter at 18kHz. this was done to help with aliasing issues before sampling, and a bessell filter was chosen to not have any ripples in the time domain, which would make subsequent lookup table creation rather tricky. sampling was done at 96kpsps and saved as 32b floats, .wav format. each sample is around 10s long.

next step is use python to apply a much steeper digital filter at 18kHz, and then slice and overlay each wave cycle onto a single waveform, then filter this and run an interpolator to get 1024 samples per cycle. the overlay process will essentially act as a synchronous oversampler, giving me much finer time based resolution than a single cycle ever could. if i use the full 10s sample in the process, it will 1000x oversampling.

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Re: 303 circuit modelling

by **connortreacy** » Sun Aug 10, 2025 3:35 pm

Out of curiosity, what's the plan for using these samples once you've got them perfect? I see the sq wave is especially quirky as you run through the pitches, but what would the samples get you here? Would you squash them to a paged wavetable for playback? I'd be more tempted to try and emulate the quirkiness in the oscillator synthesis code.

<https://github.com/connortreacy/digix0x>

Re: 303 circuit modelling

by **guest** » Sun Aug 10, 2025 4:50 pm

ive been working all week on this, and feel like ive gotten nowhere. but heres the original plan:

1. take the wav files and down/up sample to 1024 samples per cycle
2. play these back as wavetables, with a wavetable per note

i spent a bunch of time filtering and up/down sampling, and its ok for most of the range. there are 2 issues im facing with this method, though.

1. 64 tables per waveshape times 2 waveshapes times 1024 samples times 28 per sample is 256kB. this is a lot of memory. the microcontroller has it, but im also consuming a lot of memory for the filter coefficients. i could go to every-other note, and it would be more reasonable, so thats not a show stopper.
2. ill still have to do interpolation of the samples to get the distortion out of playback, which is 2 fetches and 2 multiplies. again, there are plenty of resources here, so not a show stopper. but it still seems nonideal.

im working on a solution to this, at least for the sawtooth, which i think should work fine and take next to no memory and not require interpolation. not

guest

Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am

connortreacy

Learning to Wiggle

Posts: 7

Joined: Wed Jul 09, 2025 8:23 am

Location: London

Contact: [📧](#)



guest

Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am

connortreacy

Learning to Wiggle

Posts: 7

Joined: Wed Jul 09, 2025 8:23 am

Location: London

Contact: [📧](#)



guest

Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am



guest

Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am

sure if it will work for the square.

the code as i have it does not do any antialiasing filtering on the saw, and i can hear it at the high end, and its not good. so some solution is required, and the current implementation is essentially a lookup table of an ideal sawtooth. i havent tried playing my square code through the filter yet, so i dont know how that would sound, but it would also need antialiasing as there is a very sharp falling edge. the rising edge is probably ok. my main concern with running the square code is resources. it takes a fair bit of time to calculate samples for the square, and this would go up once antialiasing had been added (some sort of polyblep or rounding of the sharp edges). its well within what the feather m4 can do, so perhaps im just being picky, but id like to see if i can get it running on an M0+ at some point, or at least an M3.

i might just be lost in the weeds, but i think i see a light at the end of the tunnel and im going to give it another day before taking the brute force approach.

hows your code coming along?

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Re: 303 circuit modelling

by [guest](#) » Sun Aug 10, 2025 9:19 pm

ok, i got something working which i think will be way less memory and slightly less processing time. i have it working for the sawtooth, and need to see if it applies to the square. basically, i have a single lookup table (right now ~4k values) which only has the saw reset waveform. this is where all the high harmonics are "stored", and the rest is just a ramp of varying slope based on note. so i run the ramp based on a calculation (which i would have to do anyways to find the appropriate sample to pick - just an incremented counter) and when i get to the transition region i subtract a value from the lookup table. this is far fewer lookup table accesses, has no FM from misaligned lookup table points (if you dont do interpolation), and has no aliasing issues. the sawtooth out of this is way cleaner than what i was getting from my lookup tables previously.

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Re: 303 circuit modelling

by [brer_rabbit](#) » Sun Aug 10, 2025 10:46 pm

I was re-reading the thread to see if you had slides done 'cause i was wondering if lookup tables versus a VCO algorithm would effect that. Then I see you got that back in February (woot!). I don't know the internals on slide but sounds like the slide solution is agnostic to your VCO implementation?

Re: 303 circuit modelling

by [guest](#) » Sun Aug 10, 2025 11:47 pm

depending upon how the wavetable is done, slides could be a problem. but, there is also a way to do it so its not, although this hybrid approach mostly sidesteps the problem altogether.

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Re: 303 circuit modelling

by [stephen](#) » Mon Aug 11, 2025 7:56 am

“ [guest](#) wrote: ☺

Sun Aug 10, 2025 9:19 pm

ok, i got something working which i think will be way less memory and slightly less processing time. i have it working for the sawtooth, and need to see if it applies to the square. basically, i have a single lookup table (right now ~4k values) which only has the saw reset waveform. this is where all the high harmonics are "stored", and the rest is just a ramp of varying slope based on note. so i run the ramp based on a calculation (which i would have to do anyways to find the appropriate sample to pick - just an incremented counter) and when i get to the transition region i subtract a value from the lookup table. this is far fewer lookup table accesses, has no FM from misaligned lookup table points (if you dont do interpolation), and has no aliasing issues. the sawtooth out of this is way cleaner than what i was getting from my lookup tables previously.

This sounds like a custom kind of BLEP algorithm, but using a sample instead of calculating the anti alias waveform, if I understand this right

(they/them)

i'm making Eurorack and 5U stuff: <https://www.gauche-electrosound.net>

new stuff: tiny pieces
harsh noise vaporwall
12 Sep 2025
[tiny pieces](#)

Re: 303 circuit modelling

by [guest](#) » Mon Aug 11, 2025 12:45 pm

yeah, thats a good way to think of it. its sort of a piecewise BLEP, where i dont bother running the algorithm when i know i dont need it (ramp) and precalculating it for the section that does. one thing i like about doing it this way, is that i can control the harmonics directly through filtering rather than in the time domain. and its my understanding the BLEP will only do smoothing, whereas this filtering technique allows for ripples (which is required for sharper filtering - a square minus higher harmonics is a rippled square).

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[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



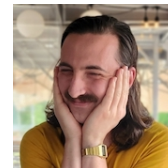
[brer_rabbit](#)
Wiggling with Experience

Posts: 251
Joined: Wed Nov 10, 2021 12:25 pm
Location: Seattle
Contact: [☺](#)



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



[stephen](#)
Veteran Wiggler

Posts: 709
Joined: Sun Jun 07, 2020 10:41 am
Location: Indianapolis, USA
Contact: [☺](#)



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

Re: 303 circuit modellingby **PrimateSynthesis** » Mon Aug 11, 2025 1:46 pm

“ guest wrote: “

Thu Nov 21, 2024 5:50 pm

yeah, it makes no sense for a bass machine, but i think its one of the things that makes it a good acid machine. the higher frequencies can be quite piercing.

What is the slope of the VCA filter? (I'm too lazy to look for my big binder of Roland service notes.) I have two TB-303's. One has the Kenton mod, and I can get good bass with the CV input. The other has a bit of a brighter sound, but it still produces decent bass pitching it down in track mode.

FS: Rare Morley Rotating Wah Oil Can Delay!

Since the forum has gone back to being 100% community funded, I've decided to donate half the proceeds from the Morley Rotating Wah this thread is supposedly selling

<https://www.muffwiggler.com/forum/viewtopic.php?t=131332>**Re: 303 circuit modelling**by **guest** » Mon Aug 11, 2025 2:00 pm

is this the filter going into the VCA or at its output? there are a bunch of single pole highpass filters sprinkled throught the circuit, and these really cut the bass end way off. if you compare your 303s, and one has the bass drop off as you turn up the resonance more so than the other one does (they all lose low end with resonance), then the offending highpass is in the filter feedback circuit. you can listen to the filter output individually on them to see if its happening there as well. the audio also goes to a mixer circuit and the headphone amp, both of which could be causing the low end loss.

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Re: 303 circuit modellingby **Electric Druid** » Mon Aug 11, 2025 4:26 pm

“ stephen wrote: “

Mon Aug 11, 2025 7:56 am

“ guest wrote: “

Sun Aug 10, 2025 9:19 pm

ok, i got something working which i think will be way less memory and slightly less processing time. i have it working for the sawtooth, and need to see if it applies to the square. basically, i have a single lookup table (right now ~4k values) which only has the saw reset waveform. this is where all the high harmonics are "stored", and the rest is just a ramp of varying slope based on note. so i run the ramp based on a calculation (which i would have to do anyways to find the appropriate sample to pick - just an incremented counter) and when i get to the transition region i subtract a value from the lookup table. this is far fewer lookup table accesses, has no FM from misaligned lookup table points (if you dont do interpolation), and has no aliasing issues. the sawtooth out of this is way cleaner than what i was getting from my lookup tables previously.

This sounds like a custom kind of BLEP algorithm, but using a sample instead of calculating the anti alias waveform, if I understand this right

Not even that "custom", that's pretty much a direct BLEP implementation. There are different ways to think about it and organise it, but it generally amounts to the same thing. For example, you can have one massive table like this which you then step through in big increments, effectively selecting different sub-groups of samples from within it. Or alternatively you can think of it as arranged as many different smaller tables based on the required sub-sample position. Sometimes one or the other way is easier depending on your platform and available resources - especially where these are limited or very limited.

Re: 303 circuit modellingby **guest** » Mon Aug 11, 2025 6:38 pm

what im doing is pretty much identical to this project:

https://mitxela.com/projects/bleps_via_state_machine

and there is a good writeup on the pros/cons vs standard blep

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**PrimateSynthesis**
Super Deluxe Wiggler

Posts: 5246

Joined: Tue Mar 15, 2011 6:50 pm

**guest**
Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am

**Electric Druid**
Veteran Wiggler

Posts: 570

Joined: Sat Mar 31, 2012 1:20 pm

Location: Portugal

Contact: [✉](#)**guest**
Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am